

Column Bounds for the Circuit Imbalance Measure

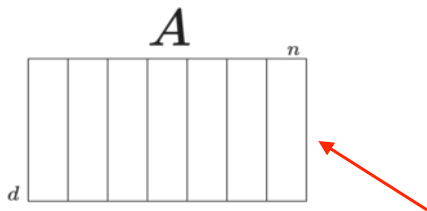
SODA 2026

Daniel Dadush, Friedrich Eisenbrand, Rom Pinchasi, Thomas Rothvoss, and **Neta Singer**.

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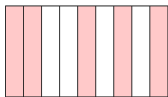


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Complexity Measures for Matrices

- Δ -modular matrices:



$$\det \left(\begin{array}{|c|c|c|c|c|} \hline \text{red} & \text{red} & \text{red} & \text{red} & \text{red} \\ \hline \end{array} \right) \leq \Delta$$

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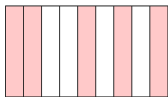
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Matrix type	Large d bounds		Large Δ bounds
TU Matrix	$\frac{1}{2}(d^2 + d)$		
Δ -modular	$\Delta^{\log_2 \log_2 \Delta + 2} d^2$	[GWZ'18]	$O(d^4 \Delta)$ [AS'23]
	$d^2 + f(\Delta)d$	[GNW'21]	$\frac{(d^2 + d)\Delta^2}{2}$ [LPSX'23]
Δ -modular (non-parallel)	$\binom{d}{2} + 80\Delta^7 d$	[PSWX'24]	

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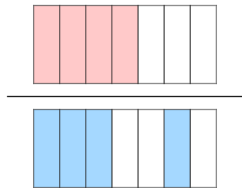
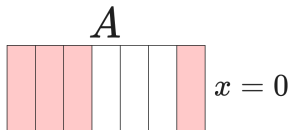
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- **Circuit imbalance measure:**

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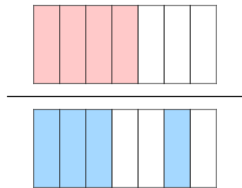
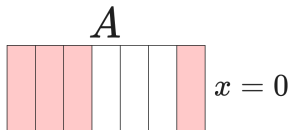


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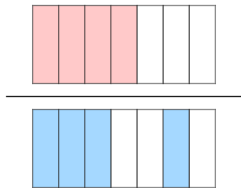
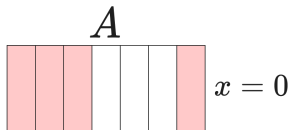
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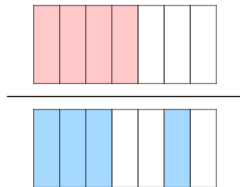
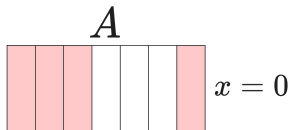
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- κ_A used to bound
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- **Question:** How many non-parallel columns can such a κ -bounded matrix have?

Our Results

Theorem (DEPRS 2026)

For $A \in \mathbb{R}^{d \times n}$ a matrix with non-collinear columns, $n \leq \pi d^4 \kappa_A$.

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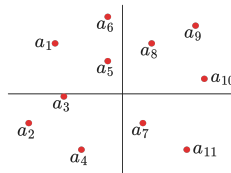
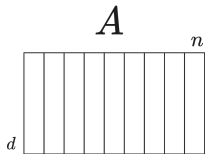
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Let $g_\infty(A)$ be the maximum entry of any Graver basis element of A , then $\|x_{IP} - x_{LP}\|_\infty \leq O(d^4 g_\infty(A)^4)$.

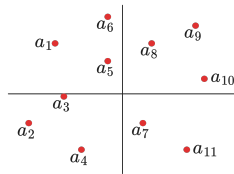
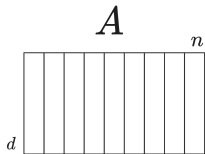
Reduction to hyperplane arrangements

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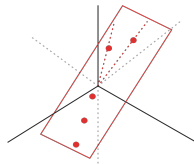
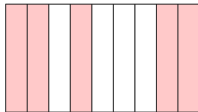


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- A_B a basis of A , $\text{span}(A_{B \setminus \{i\}})$ hyperplane.



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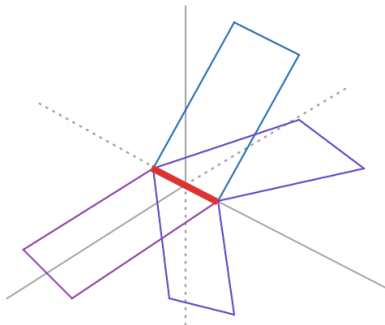
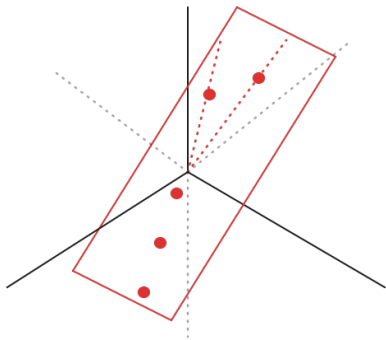
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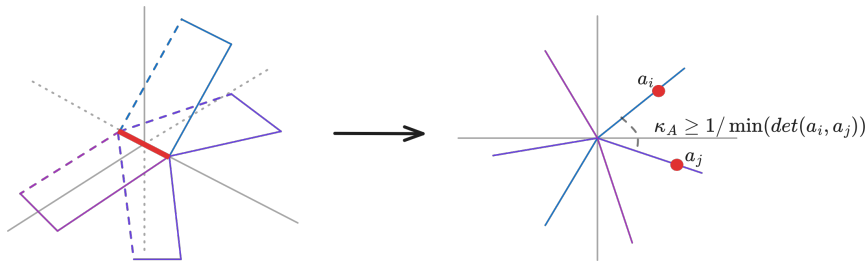
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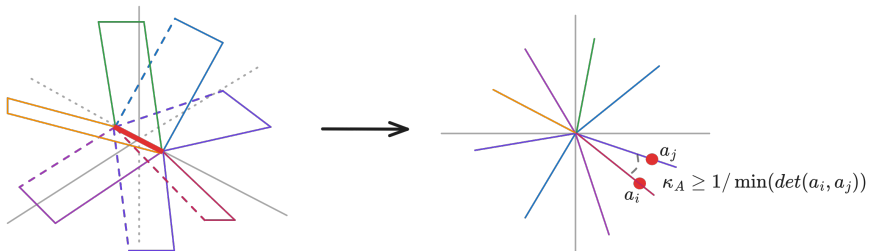
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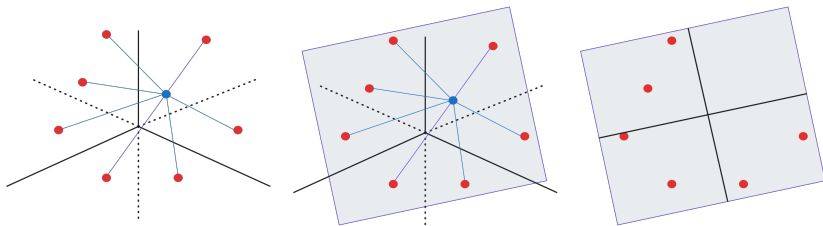
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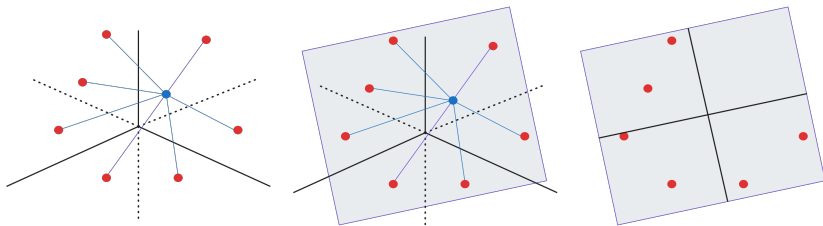
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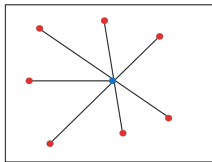
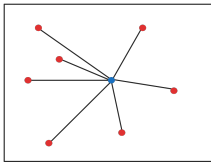
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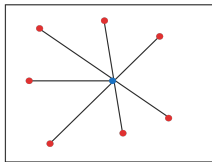
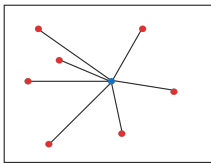
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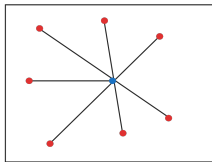
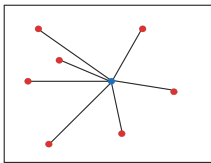
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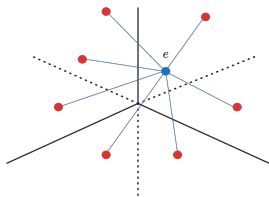
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- For full-dimensional $A \subset \mathbb{R}^d$, $\text{maxlines}(A) \geq (1 - \frac{4}{d+1})n$ [DSW'14], [DGOS'18].
- By applying the induction, the number of hyperplanes is $\geq \frac{4}{d^4} n$.

In Matroid Terminology

- Let $\mathcal{M} = (A, \mathcal{I})$ be a (complex)-representable affine matroid.
- A *circuit* of size 3 in $\mathcal{M}$ is a set of 3 collinear points.
- A *central projection* with center $e \in A$ is a contraction and simplification $si(\mathcal{M}/e)$.
- The *size* of $si(\mathcal{M}/e)$ is at least $\maxlines(A)$.
- A rank d matroid has $\maxlines(A) \geq (1 - \frac{4}{d+1})n$.

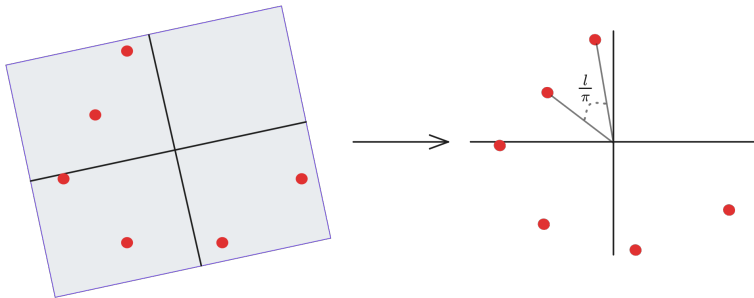


Lemma (DEPRS 2026)

A simple rank d complex representable matroid excluding $U_{2,l}$ as a minor has at most $\frac{1}{4}d^4l$ elements.

2-D base case for κ

- Contraction and simplification only decreases κ .
- Lower bounds on κ for a rank 2 matroid of size l carry over to lower bounds over $\mathcal{M}$.
- Finally, $\pi d^4 \kappa \geq n$.



Open Questions

1. Can the constant $(1 - \frac{4}{d+1})$ for the $\text{maxlines}(S)$ estimate be improved?
2. Are there other excluded minors in terms of κ ?

Thank you!

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